CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 3: Proportional Reasoning – 2

Topic-wise short notes with solved examples

Map scales, ratios with more than two terms, dividing a whole into several parts, pie charts, inverse proportion, and problems on work done together.

1 Recap of proportion

- Two quantities are in **direct proportion** when they change by the same factor. Their quotient stays constant: $\frac{x_1}{y_1} = \frac{x_2}{y_2} = k$.
- $a : b :: c : d$ exactly when $a \times d = b \times c$.
- Rule of Three: if $a : b :: c : d$ then $d = \frac{bc}{a}$.

Example 1.1 One cook mixes 6 cups of rice with 3 cups of dal, and another mixes 4 cups of rice with 2 cups of dal. Are the mixtures the same?

SOLUTION

$6 \times 2 = 12$ and $3 \times 4 = 12$

Yes, both are $2 : 1$

2 Ratios in maps

- The **Representative Fraction (RF)** of a map is the ratio of a distance on the map to the actual distance on the ground, in the same unit.
- RF $1 : 60{,}00{,}000$ means 1 cm on the map stands for 60,00,000 cm, which is 60 km.
- Useful conversion: 1 km $= 1000$ m $= 1{,}00{,}000$ cm.
- A map gives the straight-line distance, not the road distance.

Example 2.1 A map has RF $1 : 25{,}00{,}000$. Two towns are 6.4 cm apart on the map. Find the actual distance.

SOLUTION

1 cm stands for 25,00,000 cm $= 25$ km.

6.4×25

160 km

Example 2.2 A classroom 8 m long is drawn to the scale $1 : 50$. How long is it in the drawing?

SOLUTION

$800 \text{ cm} \div 50$

16 cm

3 Ratios with more than two terms

- A ratio $a : b : c : d$ says that for every a units of the first quantity there are b of the second, c of the third and d of the fourth.
- Two such ratios are proportional, $a : b : c : d :: p : q : r : s$, when $\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{d}{s}$.
- Simplest form: divide every term by the HCF of all the terms.
- If one term is known, find the value of **one part** and multiply.

Example 3.1 A spice mix uses coriander, chillies, dal and fenugreek in the ratio $8 : 4 : 2 : 1$. If only 2 chillies are available, how much of each of the others is needed?

SOLUTION

Chillies drop from 4 to 2, a factor of $\frac{1}{2}$. Apply it to every term.

4 spoons of coriander, 1 spoon of dal and half a spoon of fenugreek

Example 3.2 Red, blue and white paint are mixed in the ratio $2 : 3 : 5$. With 10 litres of white, how much red and blue are needed, and how much paint is made?

SOLUTION

5 parts $= 10$ litres, so 1 part $= 2$ litres.

4 litres of red and 6 litres of blue; 20 litres in all

Example 3.3 A library has Odia, Hindi and English books in the ratio $3 : 2 : 1$. There are 288 Odia books. Find the others.

SOLUTION

3 parts $= 288$, so 1 part $= 96$.

192 Hindi books and 96 English books

4 Dividing a whole in a given ratio

- To divide x in the ratio $a : b : c : \ldots$, the parts are

$$\frac{a}{a+b+c+\ldots} \times x, \quad \frac{b}{a+b+c+\ldots} \times x, \quad \frac{c}{a+b+c+\ldots} \times x, \ \ldots$$

Example 4.1 Concrete is made from cement, sand and gravel in the ratio $1 : 1.5 : 3$. How much of each is needed for 110 units of concrete?

SOLUTION

Total parts $= 5.5$, and $110 \div 5.5 = 20$.

20 units of cement, 30 of sand and 60 of gravel

Example 4.2 The angles of a triangle are in the ratio $1 : 3 : 5$. Find them.

SOLUTION

The sum is 180° and there are 9 parts, so one part is 20°.

$20^\circ, 60^\circ, 100^\circ$

Example 4.3 A 150-minute practice session is split among warm-up, batting, bowling and fielding in the ratio $3 : 4 : 3 : 5$. Find the time for each.

SOLUTION

Total parts $= 15$; one part $= 10$ minutes.

30, 40, 30 and 50 minutes

Example 4.4 There are 100 coins of ₹10, ₹5, ₹2 and ₹1 in the ratio $4 : 3 : 2 : 1$. Find the total value.

SOLUTION

The numbers of coins are 40, 30, 20 and 10.

Value $= 400 + 150 + 40 + 10$

₹600

Example 4.5 Can a triangle have sides in the ratio $1 : 3 : 5$?

SOLUTION

The sides would be k, $3k$, $5k$, and $k + 3k = 4k$ is less than $5k$.

No. The sum of two sides must be greater than the third.

5 Pie charts

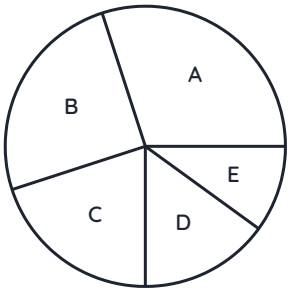
- A **pie chart** shows how a whole is divided into parts. Each part is a sector (slice) of a circle.
- The angle of a sector is proportional to the quantity it stands for. The angles add up to 360°.

$$\text{Angle of a sector} = \frac{\text{value of the part}}{\text{total value}} \times 360^\circ$$

- If the data are percentages: angle $=$ percentage $\times\ 3.6^\circ$.

- **To draw**: draw a circle and one radius; measure the first angle from it and draw the next radius; continue from each new radius; label the sectors.

Grade	A	B	C	D	E	Total
Students	12	10	8	6	4	40
Angle	108°	90°	72°	54°	36°	360°



Pie chart of grades: each student is 9°

Example 5.1 360 people chose a favourite season: 90 summer, 120 rainy and the rest winter. Find the angles for a pie chart.

SOLUTION

Winter $= 360 - 210 = 150$ people. Here each person is 1°.

Summer 90°, rainy 120°, winter 150°

Example 5.2 Favourite TV channels: entertainment 50%, sports 25%, news 15%, information 10%. Find the angles.

SOLUTION

Multiply each percentage by 3.6°.

$180^\circ, 90^\circ, 54^\circ, 36^\circ$

Example 5.3 In a pie chart the sector for "car" has an angle of 60° and stands for 18 children. How many children were surveyed? How many does a 90° sector stand for?

SOLUTION

60° is $\frac{1}{6}$ of the circle, so the total is $18 \times 6 = 108$.

90° is $\frac{1}{4}$ of 108.

108 children; 27 children

6 Inverse proportion

- Two quantities are in **inverse proportion** when one is multiplied by a factor n and the other is multiplied by $\frac{1}{n}$. Their **product stays constant**: $xy = k$.
- If (x_1, y_1) and (x_2, y_2) are two pairs of values, then $x_1 y_1 = x_2 y_2$, and so $\frac{x_1}{x_2} = \frac{y_2}{y_1}$.

Speed (km/h)	5	15	30	60
Time (hours)	18	6	3	1.5
Speed × time	90	90	90	90

- Examples: speed and time for a fixed distance; number of workers and days for a fixed job; number of pumps and time to fill a tank; number of people and days a stock of food lasts.
- **Direct or inverse?** Ask: if one quantity doubles, does the other double (direct) or halve (inverse)?
- These problems assume that every worker, pump or machine works at the same steady rate.

Example 6.1 Are x and y in inverse proportion? x: 40, 80, 25, 16 and y: 20, 10, 32, 50.

SOLUTION

The products are $800, 800, 800, 800$.

Yes

Example 6.2 20 workers lay a road in 4 days. How long will 10 workers take?

SOLUTION

$20 \times 4 = 10 \times d$

8 days

Example 6.3 2 pumps fill a tank in 18 hours. How long do 4 pumps take?

SOLUTION

$2 \times 18 = 4 \times t$

9 hours

Example 6.4 Food for 80 students lasts 15 days. If 20 more students join, how long will it last?

SOLUTION

$80 \times 15 = 100 \times d$

12 days

Example 6.5 42 machines make a batch of toys in 63 days. How many machines are needed to make it in 54 days?

SOLUTION

$42 \times 63 = m \times 54$

49 machines

Example 6.6 A car takes 2 hours at 60 km/h. How long does it take at 80 km/h?

SOLUTION

$60 \times 2 = 80 \times t$

1.5 hours

7 Working together

- Take the whole job as **1 unit of work**. If a person finishes it in t hours, the work done in 1 hour is $\frac{1}{t}$.
- Add the rates of all the workers to get the work done together in 1 hour. The time taken together is the reciprocal of that sum.
- If two workers take a and b hours alone, together they take $\frac{ab}{a+b}$ hours.

Example 7.1 Ram cuts a basket of vegetables in 1 hour and Shyam takes 1.5 hours. How long do they take together?

SOLUTION

Rates: 1 and $\frac{1}{1.5} = \frac{2}{3}$ of the job per hour. Together: $\frac{5}{3}$ per hour.

$\frac{3}{5}$ hour, that is 36 minutes

Example 7.2 A small pump fills a tank in 3 hours and a large pump in 2 hours. How long do both take together?

SOLUTION

Rates: $\frac{1}{3} + \frac{1}{2} = \frac{5}{6}$ of the tank per hour

$\frac{6}{5}$ hours, that is 1 hour 12 minutes

8 Common mistakes

- Using the Rule of Three for an inverse proportion. More workers mean **fewer** days.
- Treating every "more means more" situation as a proportion. Check that the factor is the same.
- Forgetting to change centimetres to kilometres when reading a map.
- Dividing by the number of terms instead of the **sum** of the terms when sharing in a ratio.
- Drawing pie-chart angles that do not add up to $360°$.
- Adding the times when two people work together. Add the **rates**.

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